

3.2.7. Student Data Analysis and Operations

12:38

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

Code:

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
# 1. Print all student details
```

```
print("All student Details:\n",a)
```

```
# 2. print total students
```

```
print("Total Students:", len(a))
```

```
# 3. Print all student Roll numbers
```

```
print("All Student Roll Nos", a[:,0] )
```

```
# 4. Print subject 1 marks
```

```
print("Subject 1 Marks", a[:,1])
```

```
# 5. print minimum marks of Subject 2
```

```
print("Min marks in Subject 2", a[:,2].min() )
```

```
# 6. print maximum marks of Subject 3
```

```
print("Max marks in Subject 3", a[:,3].max() )
```

7. Print All subject marks

```
a1 = np.delete(a,0,1)
```

```
print("All subject marks:", a1 )
```

8. print Total marks of students

```
print("Total Marks", a[:,1]+a[:,2]+a[:,3] )
```

9. print average marks of each student

```
a2 = np.array(a[:,1]+a[:,2]+a[:,3])
```

```
A = np.around(a2/3,1)
```

```
print(A)
```

10. print average mass of each subject

```
s1 = (a[:,1]).mean()
```

```
s2 = (a[:,2]).mean()
```

```
s3 = (a[:,3]).mean()
```

```
A1 = np.array((s1,s2,s3))
```

```
print("Average Marks of each subject", A1 )
```

11. print average marks of S1 and S2

```
A2 = np.array((s1,s2))
```

```
print("Average Marks of S1 and S2", A2 )
```

12. print average marks of S1 and S3

```
A3 = np.array((s1,s3))
```

```
print("Average Marks of S1 and S3", A3 )
```

13. print Roll number who got maximum marks in Subject 3

```
i = (a[:,3]).argmax()
```

```
r = a[i][0]
```

```
print("Roll no who got maximum marks in Subject 3", r )
```

14. print Roll number who got minimum marks in Subject 2

```
l = (a[:,2]).argmin()
```

```
R = a[l][0]
```

```
print("Roll no who got minimum marks in Subject 2", R )
```

15. print Roll number who got 24 marks in Subject 2

```
# i1 = np.argwhere(a == 24)
```

```
r1 = (a[(a[:,2])== 24 ,0]).reshape(-1,1)
```

```
print("Roll no who got 24 marks in Subject 2", r1 )
```

16. print count of students who got marks in Subject 1 < 40

```
count = np.argwhere(a[:,1]<40)
```

```
c = len(count)
```

```
print("Count of students who got marks in Subject 1 < 40", c )
```

17. print count of students who got marks in Subject 2 > 90

```
count1 = np.argwhere(a[:,2]>90)
```

```
c1 = len(count1)
```

```
print("Count of students who got marks in Subject 2 > 90:", c1 )
```

18. print count of students in each subject who got marks >= 90

```
Count = [len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>=90))]
```

```
Count = np.array(Count)
```

```
print("Count of students in each subject who got marks >= 90:", Count    )
```

19. print count of subjects in which each student got marks >= 90

```
print("Roll no:", a[:,0]  )
```

```
count0 = []
```

```
# for i in a[:,0]:
```

```
#     val = 0
```

```
#     index = int(np.argwhere(a[:,0]==i))
```

```
#     if a[index][1] >= 90:
```

```
#         val = val + 1
```

```
#     elif a[index][2] >= 90:
```

```
#         val = val + 1
```

```
#     elif a[index][3] >= 90:
```

```
#         val = val +1
```

```
#     count0.append(val)
```

```
# count0 = np.array(count0)
```

```
for i in range(len(a[:,0])):
```

```
    val = 0
```

```
    if a[i][1]>=90:
```

```
        val = val + 1
```

```
        elif a[i][2]>=90:
            val = val + 1
        elif a[i][3]>=90:
            val = val + 1
        count0.append(val)
count0 = np.array(count0)

print("Count of subjects in which student got marks >= 90:", count0 )
```

20. Print S1 marks in ascending order

```
asc = np.sort(a[:,1])
print(asc)
```

21. Print S1 marks >= 50 and <= 90

```
print(a[a[:,1]>=50])
print(a[a[:,1]<=90])
```

22. Print the index position of marks 79

```
print(np.where(a[:,1]==79))
```

Average time
0.088 s
88.00 ms



Maximum time
0.088 s
88.00 ms



1 out of 1 shown test case(s) passed

Test case 1

88 ms

Debug



Expected output

Actual output

All student Details:

All student Details:

· [[301, · 67, · 77, · 88.]

· [[301, · 67, · 77, · 88.]

· [302, · 78, · 88, · 77.]

· [302, · 78, · 88, · 77.]

· [303, · 45, · 56, · 89.]

· [303, · 45, · 56, · 89.]

· [304, · 88, · 98, · 45.]

· [304, · 88, · 98, · 45.]

· [305, · 78, · 88, · 99.]

· [305, · 78, · 88, · 99.]

· [306, · 68, · 78, · 36.]

· [306, · 68, · 78, · 36.]

· [307, · 79, · 12, · 45.]

· [307, · 79, · 12, · 45.]

· [308, · 46, · 45, · 77.]

· [308, · 46, · 45, · 77.]

· [309, · 89, · 32, · 88.]

· [309, · 89, · 32, · 88.]

· [310, · 79, · 24, · 33.]

· [310, · 79, · 24, · 33.]

Total Students: 10

Total Students: 10

All Student Roll Nos: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]

All Student Roll Nos: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310.]

Subject 1 Marks: [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]

Subject 1 Marks: [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]

Min marks in Subject 2: 12.0

Min marks in Subject 2: 12.0

Max marks in Subject 3: 99.0

Max marks in Subject 3: 99.0

All subject marks: [[67, 77, 88.]

All subject marks: [[67, 77, 88.]

· [78, 88, 77.]

· [78, 88, 77.]

· [45, 56, 89.]

· [45, 56, 89.]

· [88, 98, 45.]

· [88, 98, 45.]

· [78, 88, 99.]

· [78, 88, 99.]

· [68, 78, 36.]

· [68, 78, 36.]

· [79, 12, 45.]

· [79, 12, 45.]

· [46, 45, 77.]

· [46, 45, 77.]

· [89, 32, 88.]

· [89, 32, 88.]

· [79, 24, 33.]

· [79, 24, 33.]

Total Marks: [727, 742, 100, 721, 768, 107, 126, 160, 7]

Total Marks: [727, 742, 100, 721, 768, 107, 126, 160, 7]

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